DigiPick of the month

"A different CRT lamp"

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Harshdeep Singh

Punjab

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Web Design A Complete Introduction

by Nigel Chapman and Jenny Chapman

Published by

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WIN!

Send in your entry and you could win an exciting gift just by sharing an amusing picture with a tech angle to it. The picture should have been shot by you, and should not have been published anywhere earlier. E-mail your picture with the subject **"DigiPick"** and your postal address **on or before the 20th of this month** to **digipick@thinkdigit.com.** One prize-winning picture will be published each month.

MAN-TO-MAN

Bots To Target Bots

Defense technology—we spell it with an "s" because it's mostly about America—is ridiculously advanced. Blame it on American paranoia. We learn from **http://tinyurl.com/2pw3fv** that "Arleigh Burke" AEGIS ships can track aircraft to about 180 km, and can shoot them down even when they're up to 4,000 km away, with remote ship-based missiles. That's much more than the north-south spread of this country. The AEGIS system, if we understand it properly, targets things that come at them too fast. Google "AEGIS defense ship".

We've veered off topic a little: our point is, more and more people in diverse countries—the UK and South Korea, for instance, and now America—are preparing papers on how robots (and automated systems) should behave with flesh-based beings. At **http://tinyurl.com/32b6wm** is a PDF called "Concept of Operations for Armed Autonomous Systems," the latest. John S Canning, an engineer at the Naval Surface Warfare Centre, Dahlgren Division, a weapons research and test establishment, is aiming there to dictate how possibly robots could be told to only target their ilk.

You saw this coming: how did Asimov's Three Laws of Robotics get expanded into an entire PDF? Well, for one, the Laws are simplistic. Not suitable when you have systems as advanced as today's.

The upshot of the proposal—refer the PDF if you have the patience—is that robots should decide for themselves when to annihilate the enemy, only if that enemy consists of robots. Flesh, blood and bones should be spared—or at least, a human should be consulted when it comes to that. Quoting Canning, "Let machines target other machines, and let men target men." Problem is, how can the bots distinguish?

Another proposal: "We can equip our machines with non-lethal technologies for the purpose of convincing the enemy to abandon their weapons prior to our machines destroying the weapons, and lethal weapons to kill their weapons." Kill their weapons? Canning is referring to the bots, we presume. A related proposal is that only machines should be targeted, and if humans are killed in the process, it's collateral damage. Meaning that, faced with a man wielding an AK74, a bot should only target the weapon and try to spare the soldier.

It's all very idealistic, and the problem remains: how can a bot distinguish? Not conceptually difficult: analyse the target for the presence of DNA and RNA. But are we there yet? Or, can we throw AI at the problem?

Seen from a different angle, why would one even want to distinguish? Doesn't the military *want* to kill gun-wielding soldiers? Canning's proposals are probably trying to push us towards the ideal of pure machine warfare, with no destruction of the stuff that life is made of.

People Who Changed Computing

Dan Bricklin

Dan Bricklin

VisiCalc was the first ever spreadsheet program, developed by a young man named Dan Bricklin. The importance of this development cannot be overstated: VisiCalc is considered by many to be the killer app that turned the computer from a hobbyist thing into something that could actually be used for something productive! The program was created in 1979 by Personal Software (renamed VisiCorp) for the Apple II, fuelling the use of the Apple II as a useful tool for business.

Bricklin toyed with the idea of creating a program with keyboard calculator to punch in a few numbers and circle them with the then-new invention called the mouse to derive their sum. The first version of the program had rows and columns and some arithmetic, but didn't scroll. It became an instant success for its usability both in businesses and households to automate financial management.

He was born to a family of printers, graduated from MIT in 1973, and joined Digital to create programs for typesetting systems—a fitting profession for a printer's son! He joined the MBA programme in Harvard to get the necessary training to set up his own small business. During his first year, he decided to sell his new calculating program door-to-door for people to make financial forecasts!

Over time, Bricklin has come to be considered a revolutionary figure. Excel is something we take for granted, but now you know how it began.